Quantitative developer

[edit]

Quantitative developers, sometimes called quantitative software engineers, or quantitative engineers, are computer specialists that assist, implement and maintain the quantitative models. They tend to be highly specialised language technicians that bridge the gap between software engineers and quantitative analysts. The term is also sometimes used outside the finance industry to refer to those working at the intersection of software engineering and quantitative research.

Hypothesis of non-ergodicity of financial markets

[edit]

The nonergodicity of financial markets and the time dependence of returns are central issues in modern approaches to quantitative trading. Financial markets are complex systems in which traditional assumptions, such as independence and normal distribution of returns, are frequently challenged by empirical evidence.[16][17] Thus, under the non-ergodicity hypothesis, the future returns about an investment strategy, which operates on a non-stationary system, depend on the ability of the algorithm itself to predict the future evolutions to which the system is subject. As discussed by Ole Peters in 2011, ergodicity is a crucial element in understanding economic dynamics,[18] especially in non-stationary contexts. Identifying and developing methodologies to estimate this ability represents one of the main challenges of modern quantitative trading.[19][20] In this perspective, it becomes fundamental to shift the focus from the result of individual financial operations to the individual evolutions of the system.

Operationally, this implies that clusters of trades oriented in the same direction offer little value in evaluating the strategy. On the contrary, sequences of trades with alternating buy and sell are much more significant. Since they indicate that the strategy is actually predicting a statistically significant number of evolutions of the system.